

EnergyCostCalc

Aleksandr Petrov, Payman Sipass

¹ LIACS, Leiden University, Niels Bohrweg 1, Leiden, The Netherlands
Sanjok2003@gmail.com, payman.sipassmain@gmail.com

supervised by Reyhan Danisan , Prof Dr Ir Fons Verbeek.
reyhandanisan@gmail.com & f.j.verbeek@liacs.leidenuniv.nl

As the cost of living keeps rising year after year, making sure that you know where every cent of your income is going towards becomes gradually more important. However, when the electricity bill comes at the end of the month, you only see the total cost of all the electricity your various household appliances have used. To the average person, therefore, it can be hard to visualize how much electricity and money a specific household appliance such as an oven is costing you. This project addresses this issue by having developed a tool which not only allows the user to calculate costs for specific appliances, but also tests the user on their awareness regarding these costs. This solution aims to leave a lasting impact on users, potentially influencing their energy spending habits in a positive way.

Introduction

In times when the cost of living is increasing faster than the average salary is, economic hardship is bound to occur. Consequently, an increasing number of citizens need to be frugal and rethink their monthly budgets to get by. One of the main sources of unnecessary expenses comes from a person's electricity usage. In the modern age, the amount of household appliances that are being used, ranging from electric kettles to hair dryers, can turn into invisible money sinks due to the amount of electricity they consume. Furthermore, as most people only see the total amount of money they have spent on electricity at the end of each month, it can be hard to understand which appliances are costing you the most money compared to others. It is this hardship that led us ask the following research question:

- How can we make people more aware of how much their household devices consume energy?

In the following paragraphs, we cover preexisting approaches to answering this question, our approach, and our target userbase.

Existing Solutions

There have been other approaches and products aimed at solving this very problem. The *Sense Home* app, produced by *Sense*, is a well-designed smartphone application which breaks down the costs of individual household appliances and displays those costs in easily understandable graphs and comparisons. While the *Sense Home* app excels in what it is designed to do – showing users how much they are spending per device – it lacks a feature that confronts the user on how much they **think** they are spending. This means that, while users get to know where their money is primarily going towards, they are not being challenged to think about how their expectation of the costs might differ drastically from reality. A tool which does highlight this gap can potentially leave a more lasting impact on the user, in turn influencing their future spending habits.

User Analysis

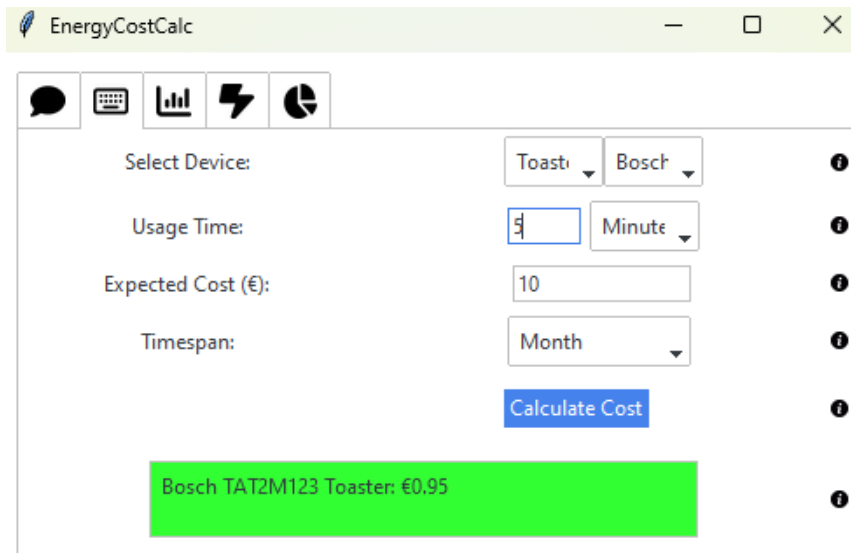
While any person who uses a variety of household appliances would benefit from a better understanding of their electricity consumption, our main target userbase consists of people of the middle, and especially lower economic classes, as they would suffer most during times of higher energy costs and overall economic hardship. Furthermore, our solution is vital for those who do not spend a lot of attention on their bills, meaning that people who are already quite aware of their electricity consumption are not part of our target userbase.

“Don’t pay more, pay attention”

Our approach to increase the awareness of electricity usage and costs is to make people come face to face with the gap between their expectations and reality. The resulting solution we came up with was to make a tool which is essentially a calculator for electricity usage and expenses. A screenshot of the tool is shown in figure 1. Using a calculator as an analogy allows users who have not interacted with the tool before to quickly grasp how to use it. The main thing that sets our tool apart, however, is that the tool asks the user to enter how much they believe the cost will be before the calculation is shown.

After the user has entered their expected cost, the tool calculates the real costs and displays graphs which highlight the difference between what the user thinks the device costs them and how much it actually costs them. By making the users input the amount of money they expect a device to cost them before we show them the actual cost, we visually highlight the gap between expectation and reality (figure 2). This, in turn, aims to leave a more lasting impression on the user, which might lead to a change in future behaviour surrounding their electricity consumption. Furthermore, after the costs are calculated and the data is shown to the user, a text box detailing possible steps to decrease these costs is presented to the user. With this, we aim to give users actionable steps to reduce their electricity bills. The tool can be run on the

Windows and Linux operating systems. By making the tool a downloadable program, it can be easily reached by many people.



The screenshot shows the 'EnergyCostCalc' application window. It features a toolbar with icons for chat, keyboard, bar chart, lightning bolt, and pie chart. The main area contains input fields for 'Select Device:' (Toaster, Bosch), 'Usage Time:' (5, Minute), 'Expected Cost (€):' (10), and 'Timespan:' (Month). A blue 'Calculate Cost' button is present. Below the inputs, a green box displays the result: 'Bosch TAT2M123 Toaster: €0.95'. Information icons are visible on the right side of the input fields.

Figure 1: The main tab of the tool which handles all user input.

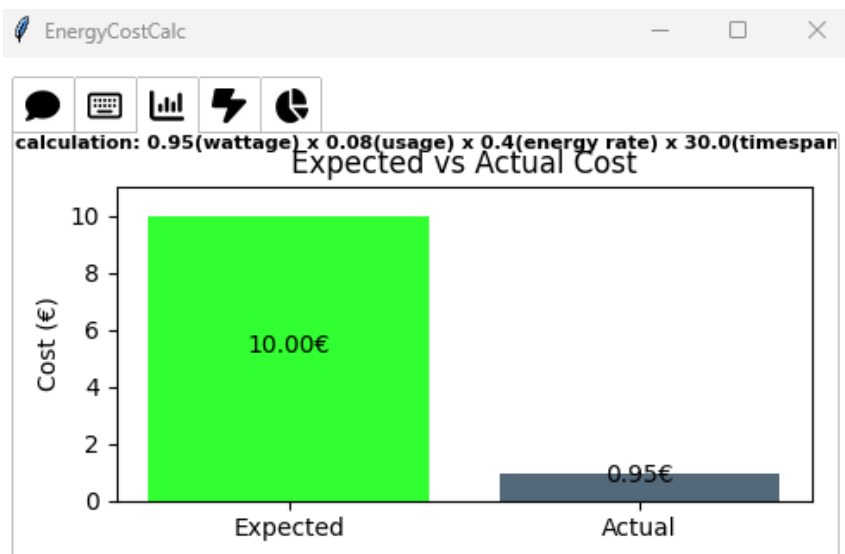


Figure 2: Bar graph highlighting the gap between the user's expectation and reality.

Usability Requirements

At the start of the project, we aimed to make the tool as clear as possible to use and understand. Consequently, we had to be meticulous about our attention to detail regarding UX and UI. Furthermore, making the tool easily accessible at any point in time was another requirement we set for ourselves. This way users could, for example, calculate potential costs of a washing machine they are thinking of buying at a store on their smartphone, instead of on their personal computer at home.

Materials and Methods

The initial methodology was strictly used which implied using Python as programming language, with its convenient libraries. The tool is written in Python3 and uses the following API's:

- *Pandas* for working with data-frames.
- *Tkinter* and *tkbootstrap* for interaction with GUI toolkit
- *Matplotlib* for data visualization
- *Pillow* for image processing.

The small database was handcrafted using a .csv file. Data collection for the specifically chosen devices and their wattage was done by searching the internet for the most sold appliances per category type (e.g., top 10 sold electric kettles) on websites such as bol.com, coolblue.com, and tweakers.com. The wattage for the “average device” option provided for each type of appliances was calculated by taking the average wattage of the aforementioned top 10 sold appliance of that type.

After processing the database using the *pandas* module, the user fills in the required information, which then gets processed by the *Tkinter* and *tkbootstrap* library in combination with self-written functions. After the data processing, the information gets displayed using the *matplotlib* library, which allows for the use of different graphs. The graphs used in our case are bar- and pie-charts. To give the tool an appealing look, the *Pillow* library makes sure that every tab is provided with a unique icon.

Results and Evaluations

To get proper feedback from real-life users other than our project assistant and fellow classmates, we conducted two rounds of evaluations with eight participants in total (four participants per round of evaluation). Both evaluations were conducted using a survey through the Google Forms platform. The following paragraphs will succinctly describe the overall results of the evaluations and the steps we took to change our prototype in response to the received feedback. This feedback was responsible for the most drastic changes to our tool. For more in depth description about all the specific

questions and answers to the survey (e.g., feedback that did not have large impact on the development), please refer to evaluation reports 1 and 2.

Our evaluations have shown that users were, on average, not accurate in guessing their electricity expenses. This shows that the assumption that most people have low awareness of their electricity expenses was indeed the case. Regarding the intuitiveness of the tool we designed, we had mixed responses. While the average intuitiveness score was a 3.6 out of 5, some participants gave a score as low as 2, while others gave a score as high as 5. This indicated that there were many components of our tool that were difficult to use or understand, which led us to revise our tool in various ways, such as including explanatory paragraphs on how the tool works behind information buttons. We received similar feedback regarding the graphs the tool produced. In addition to that, we also received comments during a project presentation regarding the visual aspects of the tool, such as the color scheme for the graphs and the icons. Consequently, this also resulted in us tweaking our tool (e.g., making the tool resizable so that the graphs are easier to read on larger screens, improving usage of colour on the graphs, etc.).

However, the most important part of our surveys was knowing whether our tool actually helped users increase their awareness of their electricity bills. This would determine whether our solution was successful in its approach to the research question. Of the eight participants who filled out our forms, seven of them reported an increase in understanding of their electricity bills, which indicates that our tool did indeed help in this regard. Some further elaboration given by the participants reveals that the tool helped them realize where they are spending the most money and that the tool has influenced their future decision-making concerning usage of various household appliances.

Limitations

As is the case with many projects, not all planned features are present in the final product. While this was mostly due to time constraints, there were also some constraints put on us by the chosen programming environment. An example of this lies with the third graph the tool produces, the pie chart. During our evaluations, it became increasingly clear that this graph was the least understood by our participants. Our solution for this issue was to add some explanatory text describing the graph next to it in a small, toggleable text box. However, this proved to be too difficult. In combination with the limited time available to work on the project, we sadly had to leave the graph as-is. Another aspect we unfortunately had to abandon was the accessibility of the tool on smartphones and macOS compatibility, as these also proved to be too time consuming to focus on. This had led to tool to only function on computers running the Windows (10 and 11) and Linux operating systems, which goes against our previously established usability requirement of being able to always access the tool easily. In the future, it may be worth exploring the idea of making this tool into a web application which could be run on any device with access to an internet browser.

Conclusion

With this project, we aimed to discover how we could increase people's awareness of how much their household devices consume energy. In response to this, we created a tool which functions not only as a calculator for your electricity expenses, but also as a program which helps you raise your awareness of how far off your expectations of those expenses are. Our main conclusion is that by highlighting this gap between prediction and reality, we were able to leave lasting impact on users, leading to them rethinking their electricity expenditure and sometimes even changing their future decision-making on the matter.

Discussion

Our answer to the research question, that being that our tool's focus on raising awareness did indeed raise awareness among users, is supported by the feedback given to us from the participants during the evaluation rounds. Of the eight participants who tested the tool and subsequently filled in the survey, seven said that they felt as if their understanding of their electricity expenses was helped by using the tool for a short period of time. Most of those participants mentioned that they appreciated the convenience of tool, allowing them to quickly figure out the cost of using a specific device, rather than just seeing the sum total of all of their devices at the end of each month. One person in particular mentioned that this straightforward insight would inform their future decision-making regarding the usage of various appliances.

Shortcomings

It would be rash to make the claim that the definitive answer to the research question is the one that we ended up with. To start with, our limited pool of testers, while helpful, is not enough to make conclusions for a wider userbase. For a more convincing end result, more test participants of different ages, living conditions, and economic classes are desirable.

Future Works

Regarding any future prospects, we aim to continue developing our tool in order to make it accessible on more platforms, turning it into a cross-platform application. Furthermore, by increasing the selection of calculatable devices we would be able to make the tool a more complete package for users wanting to calculate the energy expenses of their entire household. As previously mentioned, there also exists the possibility to create the tool in a web-based coding environment using JavaScript, circumventing the need to make platform-specific builds of the tool, as any device capable of running web-apps would be able to access it.

Acknowledgements

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